

ABSTRACT



Hybrid touch rendering coordinates a Kinova Gen3 with interchangeable feature and material modules to reproduce fingertip-scale contacts for heterogeneous virtual objects. Quest tracking yields a virtual contact frame; online feature selection and pose-QP control place a compatible physical contact beneath the finger, while continuous-surface validation governs transitions. Parallel control reached 46–49 Hz, with continuous-surface trials achieving 94.6% virtual and 93.0% real tracking success, demonstrating unified shape-and-material rendering in one non-wearable platform.

INTRODUCTION

- Encountered-type displays usually reproduce a narrow cue set or fixed geometry, limiting reuse across heterogeneous virtual objects.
- A single robot repositions interchangeable geometry and material modules to render only the fingertip-scale contact required at each instant.
- The platform therefore unifies shape and material simulation without replicating complete physical objects.

SYSTEM SETUP

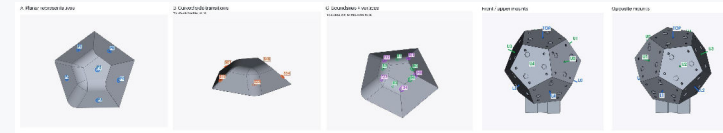


User-facing augmented-reality view and physical interaction in the integrated system.

- Quest + Unity track the fingertip and compute the virtual contact frame.
- ROS 2 selects compatible features and material faces, checks reachability, and solves the pose-QP.
- Kinova Gen3 places the selected contact beneath the fingertip; a shadow model mirrors validated commands.

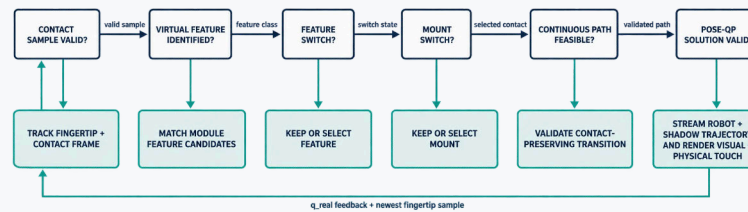
METHOD

FEATURE AND MATERIAL REPRESENTATION



- Twenty-one local feature classes encode contact position, normal, and orientation.
- Different materials occupy distinct mount faces.

REAL-TIME CONTACT-TO-ROBOT CONTROL



- Each fingertip sample drives parallel validation and command streaming; pose-QP executes same-feature motion, while feature or mount changes require validated surface paths.

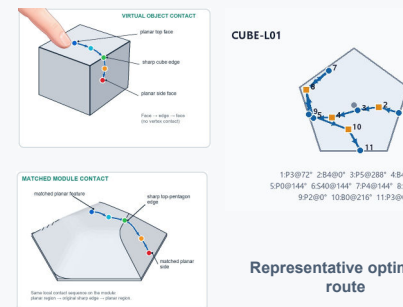
POSE-QP VELOCITY CONTROL

$$\begin{aligned} \dot{q}^* &= \arg \min \mathcal{L}(\dot{q}) \\ \dot{q} &\leq \dot{q} \leq \bar{\dot{q}} \\ \mathcal{L}(\dot{q}) &= \frac{w_p}{2} \|\mathbf{J}_p \dot{q} - \mathbf{v}_d\|_2^2 + \frac{w_\omega}{2} \|\mathbf{P}_n \mathbf{J}_\omega \dot{q} - \omega_d\|_2^2 + \frac{\lambda}{2} \|\dot{q}\|_2^2 + \frac{w_c}{2} \|\dot{q} - \dot{q}_c\|_2^2 \end{aligned}$$

Notation: $\dot{q}^*$: optimal joint velocity; $\dot{q}$, $\bar{\dot{q}}$: joint-rate bounds; $\mathbf{J}_p$, $\mathbf{J}_\omega$: translational/angular contact Jacobians; $\mathbf{v}_d$, ω_d : desired contact velocities; $\mathbf{n}$: unit contact normal; $\mathbf{P}_n = \mathbf{I} - \mathbf{n}\mathbf{n}^T$: tangent-plane projector; $\mathbf{w}_p$, $\mathbf{w}_\omega$, $\mathbf{w}_c$: position, orientation, and centering weights; λ : damping; $\dot{q}_c$: centering velocity.

FEATURE AND MODULE TRANSITION OPTIMIZATION

$$c^* = \arg \min_{c = (f, m, \psi) \in C_{\text{feas}}} [w_g E_{\text{geom}}(c) + w_r E_{\text{reach}}(c) + w_t E_{\text{trans}}(c) + w_o E_{\text{ori}}(c)]$$



Optimization Cost

- Geometric feature similarity
- Reachability from current state
- Transition distance and motion cost
- Contact-frame orientation

CONTINUOUS-SURFACE PATH OBJECTIVE

$$\gamma^* = \arg \min_{\gamma \in \mathcal{P}(f_i, f_j; \mathcal{M})} \sum_{e \in \gamma} \ell_e (1 + \eta |\kappa_e - \kappa_v|), \quad \kappa_e = \frac{\cos^{-1}(\mathbf{n}_a^T \mathbf{n}_b)}{\ell_e}$$

Notation: $\mathcal{M}$: module mesh; f_i , f_j : endpoint features; $\mathcal{P}(f_i, f_j; \mathcal{M})$: feasible surface paths; γ : candidate path; e : mesh edge; ℓ_e : edge length; η : curvature-mismatch weight; $\mathbf{n}_a$, $\mathbf{n}_b$: adjacent-face unit normals; κ_e , κ_v : module-edge and target virtual curvature.

EXPERIMENTS AND RESULTS

CONTROLLER CONFIGURATIONS

BASILINE CONTROLLER

Single-process reference pipeline without surface-continuity constraints.

PARALLEL HIGH-RATE CONTROLLER

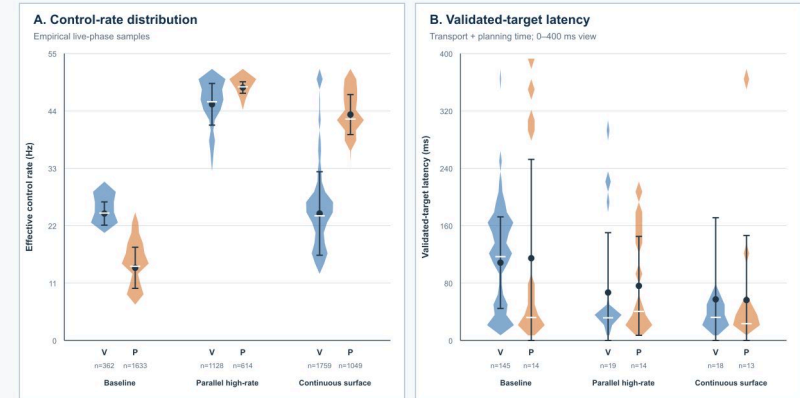
Parallelized high-rate pipeline without surface-continuity constraints.

CONTINUOUS-SURFACE HIGH-RATE

Parallelized high-rate pipeline with validated continuous feature transitions.

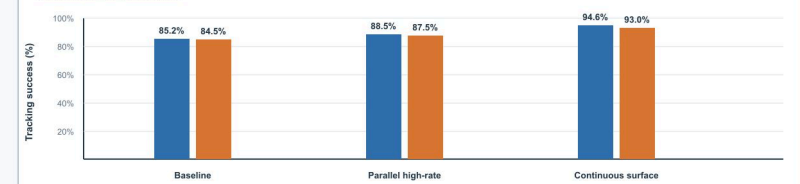
Tracking performance distributions

B = baseline - H = parallel high-rate - C = continuous-surface high-rate



C. Tracking success

Accepted target-level success rate



CONCLUSION AND FUTURE WORK

Conclusion

This work establishes a general-purpose hybrid haptic simulation framework combining robotic repositioning, feature-encoded geometry, and interchangeable materials. Thus providing a foundational solution to high fidelity non-wearable touch rendering.

Future Work

Future work will improve continuous tracking accuracy and response rate, generalize module and material selection, and add vibrotactile cues toward a general-purpose multimodal haptic simulation platform.